



Week 5 Knowledge Check

Technical Infrastructure & Modeling

Technical Product Manager Academy



How This Works

12 questions across the week — data models, cloud architecture, REST/SOAP APIs, sequence modeling, baselines & monitoring, AI-summary validation.

Commit to an answer **before** the reveal. Retrieval is the point.

? Question 1

In the "Access Pattern First" discipline, what is the *first* step before designing a schema?

- A) Pick the database engine (Postgres by default)
- B) List the queries the feature needs
- C) Draw the entity-relationship diagram
- D) Add indexes to speed up reads



Answer 1 — B

List the queries the feature needs.

Then design the schema so those queries are cheap. If a data model can't be defended by the queries that drive it, it's wrong.

? Question 2

Which storage bucket do you reach for when you need high throughput, simple lookups, caches, and sessions?

- A) Relational (Postgres, MySQL)
- B) Document (MongoDB, Firestore)
- C) Key-value (Redis, DynamoDB)
- D) Search (Elasticsearch, OpenSearch)



Answer 2 — C

Key-value (Redis, DynamoDB).

High throughput, simple lookups, caches, sessions. Relational is the B2B default; document is flexible/hierarchical; search is full-text.

? Question 3 — True / False

In this course, the TPM's job is to pick the database engine — for example, deciding Postgres vs MongoDB.

Answer 3 — False

The TPM doesn't pick the engine.

The TPM describes the access pattern clearly enough that the architect *can*.

? Question 4

In the managed-vs-self "Four-Question Test," which justification for self-managing is called a "smell"?

- A) "Failure modes don't match our needs"
- B) "Compliance forbids the managed service"
- C) "We want full control"
- D) "Cost is unreasonable at our scale"



Answer 4 — C

"We want full control."

Default to managed. Push: what *specific* decision does the managed service foreclose? The other three are legitimate reasons to self-manage.

? Question 5 — True / False

"We should be more secure" is a sufficient reason to adopt a customer-VPN-only network boundary.

Answer 5 — False

Don't volunteer for VPN-only — it **triples ops complexity**.

"More secure" is a reason for stronger TLS. Only choose customer-VPN if a customer *requires* it.

? Question 6

In the ROM cost table, which cost is called out as the *most-forgotten*?

- A) Postgres (RDS) hours
- B) Egress bandwidth
- C) S3 storage
- D) Load balancer LCUs



Answer 6 — B

Egress bandwidth.

ROM goal is rough order of magnitude. The TPM job: catch the 2x surprise before it ships — and egress is the one that hides.

? Question 7

Which HTTP status code signals an idempotency or state collision?

- A) 400 Bad request
- B) 403 Unauthorized
- C) 409 Conflict
- D) 422 Unprocessable



Answer 7 — C

409 Conflict.

401 = unauthenticated. 403 = authenticated but lacks permission. 422 = validated but violates a business rule. Specific codes are how the client behaves correctly.

? Question 8 — True / False

For a "share" action, POST /share (a verb in the URL) is the recommended REST pattern.

Answer 8 — False

URLs are nouns; the verb is the HTTP method.

POST /share is wrong. POST /shares is right.

? Question 9

In a weird-path sequence diagram, what is described as the "senior-author signal"?

- A) Showing every lifeline in the whole system
- B) Naming the invariant the design preserves
- C) Redrawing the full happy path first
- D) Adding a latency-budget note



Answer 9 — B

Naming the invariant.

"At most one ReconcileEvent per Idempotency-Key per 24h." If a triad can't name the invariant, the weird-path design isn't done.

? Question 10 — True / False

A sad-path sequence that ends with the user stuck (no recovery action) is acceptable as long as the error is rare.

Answer 10 — False

The user must have a path forward — not a dead-end.

A sad path that ends "user is stuck" is not designed. Force the recovery action.

? Question 11

The 4-Question Alert Test says that an alert which can't answer all four questions is:

- A) A page-level (Sev1) alert
- B) Noise — it'll be muted within a sprint
- C) An executive-dashboard metric
- D) A performance baseline

Answer 11 — B

Noise — it'll be muted within a sprint.

If you can't answer meaning, on-call action, threshold + window, and severity, the alert doesn't earn its page.

? Question 12

In the AI-Summary Validation Log, which status means "used as input but not yet validated — track to closure"?

- A) Cross-checked all citations
- B) Spot-checked
- C) Adopted with rationale
- D) Pending



Answer 12 — D

Pending.

Used as input but not yet validated. It needs a deadline and must be tracked to closure — otherwise it becomes "we trusted the AI."



That's Week 5

Score yourself — but the real check is whether your TMD can survive a hostile reviewer on Monday.